

Committee meeting outlines

Xiaomao Wang

September 11, 2026

1 Brief introduction to the first two chapters

A brief recap of the background and the focus of my first two chapters.

2 Chapter 1: Masting and reproductive traits

2.1 Research question

Can reproductive traits provide evidence for the evolutionary mechanisms shaping masting patterns from the different hypotheses perspective?

1. Test whether traits associated with different evolutionary hypotheses are related to the strength of masting.

2.2 Methods and analysis

1. We used phyloglm to analyse each trait separately.
 - (a) Masting was treated as a binary response variable.
 - (b) Other traits included both continuous and categorical variables.
2. We used NMDS to examine masting patterns using multiple reproductive traits.
 - (a) Test whether combinations of traits associated with a particular hypothesis are more strongly related to masting patterns.

2.3 Key results

1. Seed weight and wind pollination were the only traits associated with strong masting.
2. These associations were detected only in angiosperms.
3. We found no evidence that all traits associated with a single hypothesis were consistently associated with masting.

2.4 Alternative explanations

1. Masting is a population-level phenomenon
2. Different species selected under different hypotheses

2.5 Evolutionary perspective

1. Potential selective pressures underlying the observed associations between reproductive traits and masting.
2. Possible explanations for the different patterns observed between angiosperms and gymnosperms.

2.6 Questions to discuss

1. Alternative evolutionary perspectives for interpreting the relationship between seed weight and masting.
2. Possible explanations for the differences between angiosperms and gymnosperms.

2.7 Current status

1. Completed the outline and started writing the manuscript.
2. Plan to complete the manuscript and submit the paper by the end of this year.

3 Chapter 2: Growth-reproduction trade-off

3.1 Research Question

Is there a detectable trade-off between growth and reproduction in the context of masting?

3.2 Current Progress

1. Completed core measurements from 5 of 10 stands.
2. Still need some validation and cross-dating.
3. Developed a simple model linking growth and reproduction.

3.3 Model

1. Simple model linking growth and reproduction.
2. Currently trying to add in the climatic variables

3.4 Preliminary results

1. I plan to work more on the climate model this weekend and present preliminary results from the updated model.

4 Other activities

1. TA(?)
2. Additional fieldwork at MORA. (Tree mapping)
3. StanCon.

5 Updated timeline